

JAKE ZIONSON

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PROFESSIONAL SUMMARY

Low-latency Software Engineer with 4+ years of experience specializing in C++ and Python development within high-frequency trading and fintech environments. Proven track record in building high-throughput market data pipelines, optimizing order execution systems, and collaborating with quantitative researchers. Skilled in modern C++ (17/20), multi-threaded network programming, and Linux kernel-level optimization.

TECHNICAL SKILLS

Languages: C++ (C++17/20), Python, SQL, Bash

Low-Latency & Systems: Multithreading, Lock-free Structures, Socket Programming, Memory Management, Linux Internals

Fintech & Protocols: FIX Protocol, Market Data Feeds (ITCH/OUCH), Order Routing, Backtesting Engines

Tools & Infrastructure: Git, Docker, CMake, CI/CD, GDB, Valgrind, GoogleTest

PROFESSIONAL EXPERIENCE

Mid-Level C++ Systems Engineer — **Aetheris Financial Technologies** *London, UK | Oct 2024 – Present*

- Architected and deployed a multi-threaded C++20 market data parser, reducing tick-to-trade latency by 18% across FX and derivatives markets.
- Refactored legacy order-matching code to utilize lock-free queues, significantly reducing thread contention under high market volatility.
- Built Python-based automated testing frameworks to validate real-time execution pipelines against edge-case network conditions.

Junior C++ Developer — **\$\$\$-@@ Inc.** *London, UK | Jul 2022 – Sep 2024*

- Developed microservices in C++17 for real-time risk assessment and dynamic margin calculations within a high-throughput execution engine.
- Integrated FIX protocol interfaces for Tier-1 liquidity providers, facilitating seamless order flow management.
- Profiled application memory and CPU usage using Valgrind and perf, eliminating bottlenecks and preventing memory leaks.

EDUCATION

B.Sc. in Computer Science (First Class Honours) — **University of Bristol** *2019 – 2022*

Key Modules: Advanced Algorithms, Distributed Systems, High-Performance Computing, Systems Programming.

PROJECTS

Order Book Simulator (C++20)

- Built an in-memory limit order book capable of processing over 1 million operations per second with sub-microsecond latency.
- Implemented memory pooling and custom allocators to minimize cache misses and avoid dynamic allocation during runtime.